

CONNOR S. PICKETT, PH.D.

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EDUCATION

University of Surrey

Sep. 2022 - Aug. 2026

Ph.D. in Astrophysics

Thesis – *On the Matter of Darkness in the Local Group: Mass Modeling the Andromeda Dwarf Galaxies*

Embry-Riddle Aeronautical University - Prescott

Aug. 2017 - Dec. 2021

Bachelor of Science in Astronomy

Minor in Diplomatic & Military History

Thesis – *Changes in the Na D₁ Absorption Components of η Carinae*

RESEARCH EXPERIENCE

Mass Modeling the Andromeda Dwarf Galaxies

Sept. 2022 - Present

Dr. Michelle Collins, University of Surrey

Guildford, Surrey, UK

- Used high-resolution spectroscopy and photometry to update stellar membership and kinematics of Andromeda dwarf galaxies
- Utilized the Jeans modeling tool **GravSphere** to estimate central dark matter densities of Andromeda dwarf galaxies
- Closely collaborated with Surrey's Engineering Dwarfs at Galaxy Formation's Edge (EDGE) simulators and Lina Necib's MIT group to compare mock data with observations
- Contributed results to existing database of modeled Local Volume dwarf galaxies

Sodium D Absorption in η Carinae

March 2020 - Aug. 2022

Dr. Noel Richardson, Embry-Riddle Aeronautical University - Prescott

Prescott, AZ, USA

- Normalized, reduced, and fit spectra of η Carinae collected by the Cerro Tololo Inter-American Observatory
- Analyzed new sodium absorption feature and monitored changing absorption features at varying velocities
- Investigated spectra of stellar systems near η Carinae to compare similar absorption features

PUBLICATIONS

It's Not Just Star Formation: A trend of low dark matter densities in the Andromeda dwarf galaxy system

C.S. Pickett, M.L.M. Collins, J.I. Read, et al.

Submitted to MNRAS

Mass Modeling the Andromeda Dwarf Galaxies: Andromeda VI and Andromeda XXIII

C.S. Pickett, M.L.M. Collins, J.I. Read, et al.

MNRAS, Volume 540, Issue 2 (June 2025)

The Long-term Spectral Changes of Eta Carinae: Are they Caused by a Dissipating Occulter as Indicated by CMFGEN Models?

A. Damineli, T. R. Gull, N. D. Richardson, **C. S. Pickett**, et al.

ApJ, Volume 954, Issue 1 (September 2023)

The orbital kinematics of η Carinae over three periastra with a possible detection of the elusive secondary's motion

E. Strawn, N. D. Richardson, **C. S. Pickett**, et al.

MNRAS, Volume 519, Issue 4 (January 2023)

Changes in the Na D₁ Absorption Components of η Carinae Provide Clues on the Location of the Dissipating Central Occulter

C.S. Pickett, N.D. Richardson, T.R. Gull, et al.

ApJ, Volume 937, Issue 2 (October 2022)

Binary Parameters for the Massive Eclipsing Binary CC Cassiopeiae

N.D. Richardson, H. Pablo, **C.S. Pickett**, et al.

TESS Science Conference 2 (July 2021)

PRESENTATIONS

University of Hull - Milne Center for Astronomy 10th Anniversary Celebration <i>Mass Modeling the Andromeda Dwarf Galaxies</i> (Contributed Talk)	January 2026
Embry-Riddle Aeronautical University - Prescott - Speaker Series <i>Mass Modeling the Andromeda Dwarf Galaxies</i> (Invited Talk)	October 2025
University of California - Los Angeles <i>Mass Modeling the Andromeda Dwarf Galaxies</i> (Invited Talk)	October 2025
IAU 403 Symposium - The Hidden Beauty of the Galactic Outskirts <i>Mass Modeling the Andromeda Dwarf Galaxies</i> (Virtual Poster)	October 2025
Dartmouth University - Galactic Frontiers II <i>Mass Modeling the Andromeda Dwarf Galaxies</i> (Poster)	June 2025
AIP Potsdam - Junge AG Joined Summer School 2025 <i>Mass Modeling the Andromeda Dwarf Galaxies</i> (Contributed Talk)	June 2025
Durham University - Small Galaxies, Cosmic Questions II <i>Mass Modeling the Andromeda Dwarf Galaxies: Andromeda VI and Andromeda XXIII</i> (Poster)	July 2024
Royal Astronomical Society <i>Mass modeling the classical Andromeda dwarfs</i> (Poster)	April 2024
Cardiff University - National Astronomy Meeting 2023 <i>Mass modeling the classical Andromeda dwarfs: Andromeda VI and Andromeda XXIII</i> (Virtual Poster)	July 2023

WORK EXPERIENCE

Residential Teacher <i>Cardigan Mountain School Summer Session</i>	June - Aug. 2022 <i>Canaan, NH, USA</i>
· Created and taught an experiential learning curriculum for introductory and intermediate Python coding courses for kids aged 6 to 15 · Helped run afternoon physical activities, including kayaking and all-camp competitions · Oversaw a dormitory environment of boys aged 8-11, alongside two other dorm parents	
Lead Student Astronomer <i>Embry-Riddle Aeronautical University</i>	Oct. 2020 - Dec. 2021 <i>Prescott, AZ, USA</i>
· Operated on-campus Meade LX600 0.4 meter telescope, along with a connected Echelle spectrograph, to collect data on various systems · Created, oversaw, and trained a team of five students on operations of the observatory	
Planetarium Employee <i>Jim and Linda Lee Planetarium</i>	Sep. 2020 - Dec. 2021 <i>Prescott, AZ, USA</i>
· Presented planetarium shows regarding the STEM field to the Embry-Riddle Aeronautical University population and Prescott community · Utilized DigiStar 6 to present in-person and livestreamed shows to a wide range of audiences	
Teaching Assistant – Physics Laboratory for Engineers <i>Embry-Riddle Aeronautical University</i>	Aug. 2020 - Dec. 2020 <i>Prescott, AZ, USA</i>
· Taught and assisted with the learning of intermediate physics concepts, such as gravitational attraction, inductance and capacitance, and geometric optics · Graded completed lab reports in a professional, constructive manner to help prepare for a professional laboratory setting	
STEM Outreach Lead Astronomer <i>Embry-Riddle Aeronautical University</i>	Sep. 2019 - Dec. 2020 <i>Prescott, AZ, USA</i>
· Oversaw a team of three students for the operation and maintenance of Celestron telescopes · Educated visitors on astronomical events and celestial objects, at which the telescopes were directed	

Head Counselor*Camp Stonewall*June - Aug. 2019
Thompson, CT, USA

- Headed up a team of three counselors and two interns for male campers aged 10 & 11
- Ran the CSW Arts & Crafts and Photography programs, as well as various afternoon sporting activities and competitions
- Worked with the CSW Leadership Team to promote a safe, fun environment for all campers

Camp Counselor*Camp Stonewall*June - Aug. 2018
Thompson, CT, USA

- One of two counselors for male campers aged 14 & 15
- Ran various afternoon sporting activities and competitions throughout the summer for campers aged 5-15

GRANTS AND AWARDS

W.M. Keck Observation Time

October 2025

- Awarded 2 nights of observations of the Cassiopeia III and Lacerta I dwarf galaxies

UKRI/STFC PhD Studentship Grant

Sep. 2022 - Present

- Received monthly PhD stipend and a £1000 yearly travel grant

Outstanding Graduate in Astronomy

December 2021

- Granted for my outstanding academic achievement, integrity, and prowess in my field and in the ERAU Astronomy Department

ERAU Undergraduate Research Institute Grant

Sep. 2020 - May 2021

- Awarded \$2500 for research during the 2020-2021 academic year

Dean's Scholarship

Aug. 2017 - Dec. 2021

- Received \$9400 each academic year for excellent academic standing within the university

LEADERSHIP SKILLS

Surrey Lacrosse Treasurer

May 2024 - June 2025

- Oversaw club finances, as well as the organization of weekly travel, for all three Surrey Lacrosse teams
- Actively liaised with the Students' Union and Team Surrey for administration- and team-level operations
- Attended multiple professional workshops to help build a stronger, more inclusive club

Department Coffee Organizer

Oct. 2023 - Dec. 2024

- Responsible for the purchase of coffee for the Surrey Maths and Physics Department and for regular maintenance communal coffee machines

Surrey Astro Group Meeting Organizer

Sep. 2023 - Sep. 2024

- Organized and led bi-weekly Surrey Astro Group Meetings and created a clearly-communicated, semester-long presenter schedule

ERAU Order of Omega Treasurer

Nov. 2020 - Apr. 2021

- Oversaw society finances for the academic year and assisted in organizing campus-wide fundraising events

ERAU Society of Physics Students Treasurer

Apr. 2019 - Apr. 2020

- Granted SPS National's 'Outstanding Chapter Award' and Blake Lilly Prize for our chapter's efforts throughout my tenure
- Oversaw club finances for the academic year, which involved the organization of multiple successful campus-wide fundraising events

OUTREACH

- World Space Week**
University of Surrey

Guildford, Surrey, UK

 - 2025 – Assisted in explanation and public use of University telescopes, in conjunction with the Guildford Astronomical Society
 - 2023 – Organized and ran a Solar System-themed escape room for younger attendees of Space Week
 - 2022 – Presented and explained binary star simulations to visiting event members
- SPS Physics Demos**
Society of Physics Students

Prescott, AZ, USA

 - Presented various physics demonstrations to a wide age range of public attendees
 - Assisted the Highlands Center for Natural History with public physics demonstrations
- Public Stargazing**
Embry-Riddle Aeronautical University

Prescott, AZ, USA

 - Presented at public stargazing shows of the solar system and Deep Sky Objects with the ERAU Astronomy and STEM Outreach Departments
 - Helped promote STEM Education within the local community through events such as school visits

TECHNICAL SKILLS

Python	<i>Advanced</i>
L ^A T _E X	<i>Advanced</i>
Jupyter Lab	<i>Intermediate</i>
MATLAB	<i>Beginner</i>
Mathematica	<i>Beginner</i>

LANGUAGES

English	<i>Native</i>
Spanish	<i>Intermediate</i>
German	<i>Beginner</i>

PROFESSIONAL MEMBERSHIPS

Royal Astronomical Society	May 2024 - Present
Sigma Pi Sigma Honor Society	Feb. 2021 - Present
Lambda Chi Alpha Fraternity	Feb. 2020 - Present

HOBBIES

Lacrosse, Stargazing, Photography, Astrophotography, Hiking, Traveling